

① CCTV image with salt-and-pepper noise

Technique: median Filter

Reason:

- Removes salt-and-pepper noise effectively
- Gives a cleaner CCTV image.

② Satellite image with Gaussian noise

Technique: Gaussian Filter

Reason:

- Reduces Gaussian noise
- Smooths the image.

③ Medical X-ray with random noise and fine edges.

Technique: median Filter.

Reason:

- Removes random noise
- Helps maintain important medical structures.

④ Scanned document with random black and white dot

Technique: median Filter.

Reason:

- Removes black and white dot noise.
- Improves document readability.

⑤ Low-light image with heavy grain and unclear boundaries:

Technique: Gaussian Smoothing + Contrast Enhancement.

Reason:

→ makes boundaries easier to identify.



⑥ Foggy road image with very low contrast.

Technique: Contrast Enhancement:

Reason:

- makes road objects more visible
- improves overall image clarity.

⑦ underwater image that looks dull and washed out

Technique: Histogram Equalization.

Reason:

- Improves image contrast
- makes the underwater scene clearer

⑧ Chest x-ray with poor visibility of internal structures

Technique: Histogram Equalization.

Reason:

- Enhances contrast
- Helps improve the clarity of the x-ray.

⑨ Night-time traffic image with low intensity variation.

Technique: Contrast Enhancement (Histogram Equalization)

Reason:

- Increases intensity variation.
- Improves visibility in dark areas.

⑩ Grayscale image using only a narrow intensity range

Technique: Contrast Stretching

Reason:

- uses a wider range of gray levels
- Improves image contrast and visibility.